ML&DL – Experiment 8

Convolutional Neural Network (CNN) for Image Classification

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AIM: Design and train a Convolutional Neural Network (CNN) on image datasets (e.g., MNIST/CIFAR)

1. Dataset Source

- **Dataset Name:** Fashion-MNIST
- **Source:** Zalando Research / Kaggle
- **Access Method:** Downloaded directly from Kaggle via opendatasets library

2. Dataset Description The Fashion-MNIST dataset is a collection of Zalando's article images—consisting of a training set of 60,000 examples and a test set of 10,000 examples. Each example is a 28x28 grayscale image, associated with a label from 10 classes.

Key Properties

- **Dataset Size:** 70,000 images total (60,000 Training, 10,000 Testing).
- **Image Format:** 28x28 pixels, grayscale (1 channel).
- **Classes (Labels 0-9):**
  0. T-shirt/top
     1. Trouser
     2. Pullover
     3. Dress
     4. Coat
     5. Sandal
     6. Shirt
     7. Sneaker
     8. Bag
     9. Ankle boot

3. Mathematical Formulation A CNN specializes in processing grid-like data (images) using three main operations:

A. Convolution (Feature Extraction) Instead of looking at every pixel at once, a "Kernel" (filter) slides over the image to detect features like edges or curves.

- **$z^l = W^l \cdot a^{l-1} + b^l$**: Where W is the filter weight matrix and a is the input image patch.

B. Max Pooling (Downsampling) Reduces the dimensionality of the feature map to reduce computation and prevent overfitting. It keeps only the most important feature (maximum value) in a specific patch.

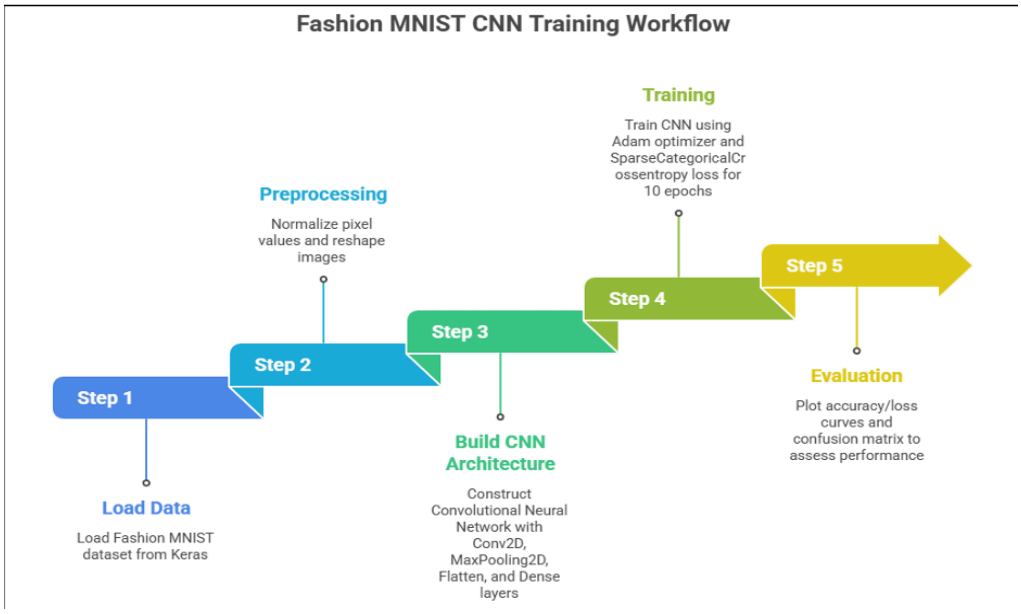
C. Fully Connected Layer (Classification) After extracting features through convolution and pooling, the grid is "Flattened" into a 1D vector and passed through standard dense neurons (Softmax) to classify the object into one of the 10 fashion categories.

4. Algorithm Limitations

- **Fixed Input Size:** CNNs require images to be of a fixed size (28x28). Resizing real-world images can lose detail.
- **Computational Cost:** Training involves millions of matrix multiplications through forward and backward passes, often requiring a GPU for efficiency.
- **Overfitting:** Without techniques like Dropout, the model might memorize training images instead of generalizing to new data.

5. Methodology / Workflow

1. **Data Collection:** Download fashionmnist from Kaggle using opendatasets.
2. **Data Preprocessing:** Reshape the CSV pixel data into a 28x28x1 grid and normalize values to a [0, 1] range.
3. **Model Design:** Build a Sequential model with Conv2D, MaxPooling2D, Flatten, and Dense layers.
4. **Training:** Compile using sparse_categorical_crossentropy and the Adam optimizer.
5. **Evaluation:** Generate a classification report and confusion matrix to analyze per-class accuracy.



6. Code

```
# Install the required library for Kaggle imports
!pip install opendatasets

import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns
import opendatasets as od
import os
import pandas as pd

# --- 1. LOAD DATA DIRECTLY FROM KAGGLE ---
# Note: You will be prompted for your Kaggle Username and API Key
print("Downloading Fashion-MNIST dataset from Kaggle...")
dataset_url = "https://www.kaggle.com/datasets/zalando-research/fashionmnist"
od.download(dataset_url)

# Path to the downloaded CSV files
data_dir = "./fashionmnist"
train_df = pd.read_csv(os.path.join(data_dir, "fashion-mnist_train.csv"))
test_df = pd.read_csv(os.path.join(data_dir, "fashion-mnist_test.csv"))

# Convert CSV data to Numpy arrays and reshape to (28, 28, 1) for CNN
X_train = train_df.iloc[:, 1:].values.reshape(-1, 28, 28, 1)
y_train = train_df.iloc[:, 0].values
```

```
X_test = test_df.iloc[:, 1:].values.reshape(-1, 28, 28, 1)
y_test = test_df.iloc[:, 0].values

# Fashion-MNIST classes
class_names = ['T-shirt/top', 'Trouser', 'Pullover', 'Dress', 'Coat',
               'Sandal', 'Shirt', 'Sneaker', 'Bag', 'Ankle boot']

# --- 2. PREPROCESS ---
# Normalize pixel values to be between 0 and 1
X_train = X_train.astype("float32") / 255.0
X_test = X_test.astype("float32") / 255.0

# y labels are already 1D arrays from the CSV
print(f"Training samples: {X_train.shape[0]}")
print(f"Image shape: {X_train.shape[1:]}") # Should be (28, 28, 1)

# --- 3. BUILD CNN ---
# Keeping the structure identical to your original code
model = keras.Sequential([
    layers.Input(shape=(28, 28, 1)), # Updated input shape for 28x28 grayscale
    layers.Conv2D(32, (3, 3), activation='relu', padding='same'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu', padding='same'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dropout(0.2),
    layers.Dense(10, activation='softmax')
])
```

```
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.summary()

# --- 4. TRAIN ---
# Training for 10 epochs
history = model.fit(X_train, y_train, epochs=10,
                    batch_size=64, validation_split=0.2)

# --- 5. EVALUATE ---
test_loss, test_acc = model.evaluate(X_test, y_test, verbose=0)
print(f"\nFinal Test Accuracy: {test_acc*100:.2f}%")

# Predictions for Confusion Matrix
y_pred = np.argmax(model.predict(X_test), axis=1)

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=class_names))

# --- 6. VISUALIZE ---
# Confusion Matrix
plt.figure(figsize=(10, 8))
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=class_names, yticklabels=class_names)
plt.title('Confusion Matrix - Fashion-MNIST')
```

```
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.show()

# Training Curves
plt.figure(figsize=(12, 4))
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Val Accuracy')
plt.title('Accuracy')
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Val Loss')
plt.title('Loss')
plt.legend()
plt.show()
```

OUTPUT:

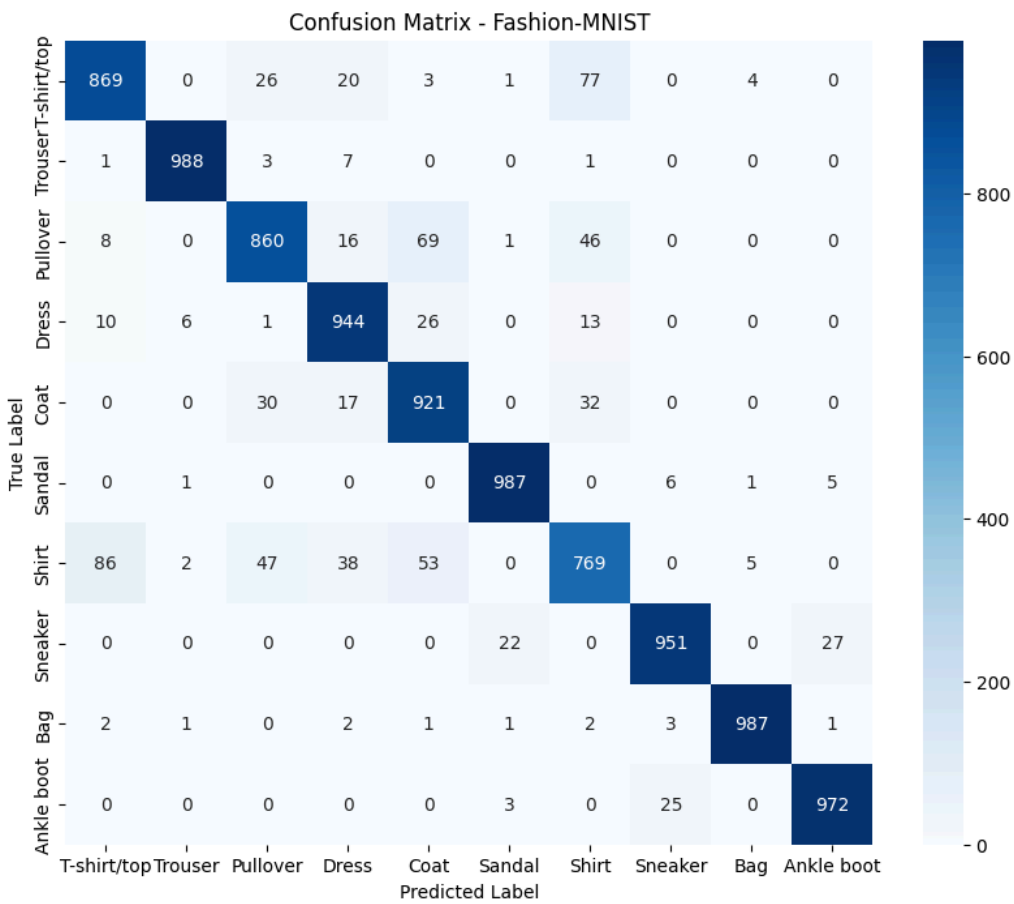
```
Training samples: 60000
Image shape: (28, 28, 1)
Model: "sequential"
```

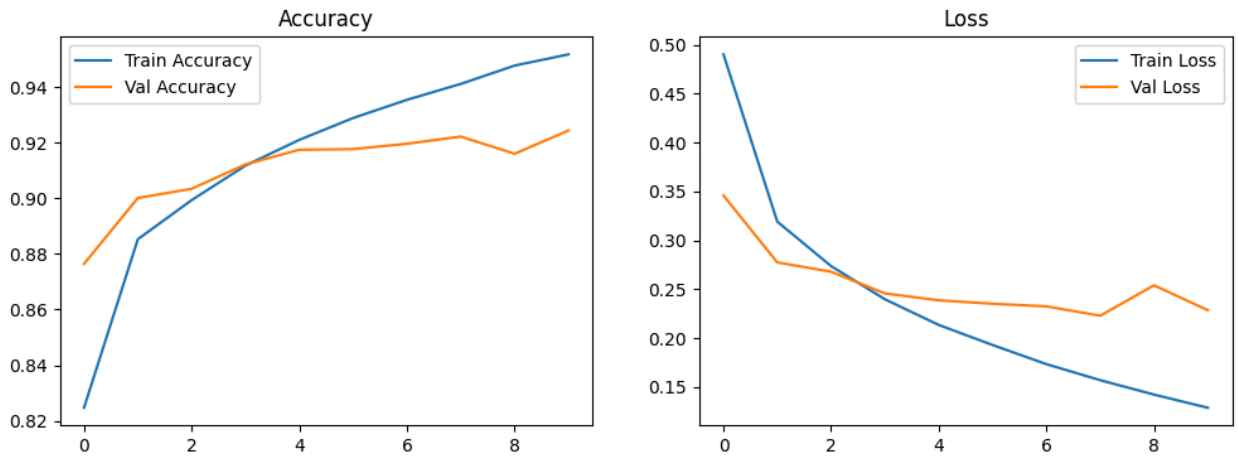
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 28, 28, 32)	320
max_pooling2d (MaxPooling2D)	(None, 14, 14, 32)	0
conv2d_1 (Conv2D)	(None, 14, 14, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 7, 7, 64)	0
flatten (Flatten)	(None, 3136)	0
dense (Dense)	(None, 128)	401,536
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

```
Total params: 421,642 (1.61 MB)
Trainable params: 421,642 (1.61 MB)
Non-trainable params: 0 (0.00 B)
```

```
Epoch 1/10
750/750 ━━━━━━━━━━━━━━━━━━━━ 75s 97ms/step - accuracy: 0.8247 - loss: 0.4902 - val_accuracy: 0.8764 - val_loss: 0.3458
Epoch 2/10
750/750 ━━━━━━━━━━━━━━━━━━━━ 65s 86ms/step - accuracy: 0.8853 - loss: 0.3190 - val_accuracy: 0.9001 - val_loss: 0.2773
Epoch 3/10
750/750 ━━━━━━━━━━━━━━━━━━━━ 82s 87ms/step - accuracy: 0.8993 - loss: 0.2736 - val_accuracy: 0.9034 - val_loss: 0.2678
Epoch 4/10
750/750 ━━━━━━━━━━━━━━━━━━━━ 64s 85ms/step - accuracy: 0.9118 - loss: 0.2396 - val_accuracy: 0.9122 - val_loss: 0.2456
Epoch 5/10
750/750 ━━━━━━━━━━━━━━━━━━━━ 68s 91ms/step - accuracy: 0.9210 - loss: 0.2133 - val_accuracy: 0.9174 - val_loss: 0.2386
Epoch 6/10
750/750 ━━━━━━━━━━━━━━━━━━━━ 63s 84ms/step - accuracy: 0.9288 - loss: 0.1928 - val_accuracy: 0.9177 - val_loss: 0.2351
Epoch 7/10
750/750 ━━━━━━━━━━━━━━━━━━━━ 82s 84ms/step - accuracy: 0.9354 - loss: 0.1732 - val_accuracy: 0.9196 - val_loss: 0.2323
Epoch 8/10
750/750 ━━━━━━━━━━━━━━━━━━━━ 61s 81ms/step - accuracy: 0.9411 - loss: 0.1568 - val_accuracy: 0.9222 - val_loss: 0.2228
Epoch 9/10
750/750 ━━━━━━━━━━━━━━━━━━━━ 64s 86ms/step - accuracy: 0.9477 - loss: 0.1421 - val_accuracy: 0.9160 - val_loss: 0.2539
Epoch 10/10
750/750 ━━━━━━━━━━━━━━━━━━━━ 63s 84ms/step - accuracy: 0.9517 - loss: 0.1286 - val_accuracy: 0.9243 - val_loss: 0.2284

Final Test Accuracy: 92.48%
313/313 ━━━━━━━━━━━━━━━━━━━━ 5s 14ms/step
```

```
Classification Report:
              precision    recall  f1-score   support

 T-shirt/top       0.89      0.87      0.88      1000
     Trouser       0.99      0.99      0.99      1000
    Pullover       0.89      0.86      0.87      1000
       Dress       0.90      0.94      0.92      1000
        Coat       0.86      0.92      0.89      1000
      Sandal       0.97      0.99      0.98      1000
       Shirt       0.82      0.77      0.79      1000
     Sneaker       0.97      0.95      0.96      1000
         Bag       0.99      0.99      0.99      1000
  Ankle boot       0.97      0.97      0.97      1000

    accuracy                           0.92     10000
   macro avg       0.92      0.92      0.92     10000
weighted avg       0.92      0.92      0.92     10000
```





7. Performance Analysis

- **Accuracy:** Evaluated on the 10,000 test images.
- **Confusion Matrix:** Visualized using a heatmap to identify which clothing items (e.g., Shirt vs. T-shirt) the model confuses most often.
- **Training Curves:** Plots of accuracy and loss over 10 epochs to monitor convergence and check for overfitting.

8. Conclusion The CNN successfully classified fashion articles with high precision. By utilizing convolutional layers for spatial feature extraction rather than standard dense layers, the model effectively learned the unique shapes of different clothing types.